

How was my performance? Exploring the role of anchoring bias in AI-assisted decision making

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ABSTRACT

Organizations leverage artificial intelligence (AI) to analyze data and support decision making. However, the integration of AI into organizational workflows may introduce unintended biases. Despite the proliferation of AI in organizations, no study to date has juxtaposed the impact of human and AI recommendations on decision making. Using two controlled experiments of 775 managers, we explore the impact of AI and cognitive bias on performance appraisal ratings. In particular, we examine anchoring and adjustment bias and present an effective strategy for mitigating this bias. The findings show managers' performance ratings are impacted by the presence of an AI recommendation. The source of the recommendation (human or AI) interacted with the anchor (high or low) to influence managers' rating. In particular, a high-anchor produced different performance ratings for each source. However, when exposed to a low-anchor, supervisors did not produce varied estimates from AI and non-AI recommendations. These findings suggest managers should be aware of the differential effects of anchoring and adjustment bias on organizational decisions. An employee's performance may be rated differently, not because of the employee's behavior, but because of the source of the recommendation and the magnitude of the anchor. This paper makes several significant contributions: (1) it is among the first studies to empirically test the presence and salience of anchoring bias in AI-assisted decision making; (2) it presents the consider-the-opposite strategy as an approach to effectively debias the anchoring effects of AI recommendations.

1. Introduction

The proliferation of artificial intelligence (AI) in organizations has transformed the way humans and machines interact. AI is often referred to as “technology that enables computers and machines to simulate human intelligence and problem-solving capabilities” (IBM, 2024). It has rapidly integrated into various sectors, reshaping traditional workflows (Berente et al., 2021). The Worldwide Artificial Intelligence Spending Guide forecasts industrial spending on AI will surge from \$85 billion in 2021 to over \$204 billion by 2026 (Business Wire, 2021). According to Gartner (2021), one in three corporations integrates artificial intelligence (AI) into their workflows to leverage the value of organizational data. It is now common for AI to assist human experts in a variety of high-stakes tasks. For instance, managers use AI recommendations in human resource management (Budhwar et al., 2022), judges use AI tools for risk assessments (McKay, 2020), and financial institutions deploy machine learning models to monitor credit risk (Alkhalili et al., 2021).

The interplay between AI and human decision makers has attracted widespread attention from both academics and practitioners, as an increasing number of managers leverage AI-generated recommendations to inform organizational decisions (Bauer et al., 2023; Dwivedi & Wang, 2022; Fügener et al., 2022; Grover et al., 2022; Jussupow et al., 2021; Nishant et al., 2024; Vazirani et al., 2023). Humans frequently fail to calibrate their dependence on AI during decision making (Dwivedi et al., 2023; Green & Chen, 2019). Overreliance on AI outputs without sufficient evaluation can lead to suboptimal decisions (Demlehner & Laumer, 2024; Ha & Kim, 2024).

One prominent cognitive bias is anchoring and adjustment, also referred to as anchoring bias (Demlehner & Laumer, 2024). Anchoring bias occurs when an initial piece of information, the “anchor,” exerts an undue influence on subsequent judgments and decisions. This effect persists even if the anchor is arbitrary or irrelevant (Tversky & Kahneman, 1974). In the context of AI-assisted decision making, anchoring bias manifests when decision makers are presented with an AI-generated recommendation as a reference point, which can skew their subsequent

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judgments and lead to biased outcomes. Anchoring introduces a reference frame in an individual's mind and associates it with human cognition such that the anchor factors into the individual's ultimate decisions (Furnham & Boo, 2011; Oz et al., 2020). Nagtegaal et al. (2020, p. 565) posit "anchoring bias is robust across institutional contexts for managers as well as employees. It should be considered when designing decision environments." The anchoring effect has been instrumental in explaining a wide range of phenomena related to organizational decision making, including expert consensus (Goldberg et al., 2019), project duration estimation (Lorko et al., 2019), salary negotiations (Hernandez et al., 2019), and quantitative evaluation of employees' performance (Nagtegaal et al., 2020).

In addition, as advancements in AI-aided decision-support systems have transformed business operations, supervisors increasingly use AI to assist employee performance management and appraisal (Ray, 2022). Despite the potential for increased efficiency, the integration of AI into these processes can introduce biases, particularly anchoring bias. For example, when evaluating a new employee, an AI system recommends a rating of 83 out of 100 based on the limited data available to the system. However, a higher rating, closer to 90, is more appropriate, reflecting that the employee consistently exceeds expectations and takes on additional responsibilities. If the manager anchors the assessment to the AI-recommended rating, the result likely undervalues the employee's actual performance. O'Connor and Liu (2024) highlight that the biases introduced by AI systems pose significant concerns for both researchers and practitioners, advocating for further investigation into the inherent biases within AI systems. Fang et al. (2024) identify substantial biases in AI-generated content, underscoring the pervasive nature of these issues. Understanding how AI recommendations influence decision making within organizations is crucial, as the biases introduced by AI can lead to unfair or inaccurate performance evaluations with far-reaching consequences (Attard-Frost et al., 2023). NIST (2022) also underscores the importance for organizations to obtain a better understanding of AI-related biases, including anchoring bias, and address them to maintain trust in AI systems and mitigate harmful impacts.

To date, the mechanism through which AI may bias performance appraisal has been underexplored. Hence, this paper addresses the following research questions: *What is the impact of the recommendation source (human or AI) and anchoring bias on decision making? If present, can the AI-based anchoring bias be mitigated through a consider-the-opposite (CTO) strategy?* Utilizing dual-process theory, our research investigates how decision makers interact with AI recommendations in the context of human performance appraisal. In particular, we explore how a recommendation source (human vs. AI) and an anchor (high vs. low) impact a manager's performance appraisal rating.

This paper makes several significant contributions. Firstly, it is one of the first studies to empirically test the presence and salience of anchoring bias in AI-assisted decision making in relation to human performance appraisal, revealing how AI-generated recommendations can influence managerial judgments, potentially leading to skewed performance evaluations. Secondly, it introduces the CTO strategy as an effective intervention to mitigate the anchoring effects of AI recommendations, demonstrating the practicality of encouraging decision makers to critically evaluate and challenge AI-generated anchors. Lastly, this research establishes a multidisciplinary foundation for further theoretical exploration at the intersection of cognitive bias, AI, and organizational decision-making. The insights garnered from this study are crucial for managers, policymakers, and AI system designers, offering them guidance on recognizing and addressing the cognitive biases introduced by AI, thereby promoting more equitable and informed decisions across various contexts.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 outlines the research design. Sections 4 and 5 detail the two experiments investigating the anchoring effect of AI recommendations on performance appraisals and assess the effectiveness of the CTO strategy as a debiasing intervention for AI-generated

anchors. Section 6 employs the Critical Incident Technique to validate the experimental findings. Finally, Sections 7 and 8 discuss the implications of our results and conclude the paper.

2. Background and hypotheses

2.1. Anchoring bias

Cognitive biases have been investigated across disciplines, including applied psychology (Berthet, 2021; LeMoult & Gotlib, 2019; White et al., 2021), marketing (Köcher et al., 2019), medicine (Lambe et al., 2016; Manning et al., 2021; Royce et al., 2019), tourism management (Royce et al., 2019), forensic science (Cooper & Meterko, 2019; Neal et al., 2022), and public administration (Battaglio Jr et al., 2019). Anchoring bias, one of the prominent cognitive biases, occurs in two stages—anchoring and adjustment (Lieder et al., 2018; Tversky & Kahneman, 1974). In Stage 1, decision makers generate a preliminary estimate on the basis of an initial piece of information, the anchor. In Stage 2, they adjust that initial estimate by reasoning or incorporating additional information. Tversky and Kahneman (1974) note that different anchors yield different estimates and that individual decisions are biased toward anchors. Additionally, adjustment is usually insufficient, further impacting the decision outcome and quality (Brandt et al., 2015).

An early experiment by Tversky and Kahneman (1974) reveals when asked to compute $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ and $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$, high school students gave very different median estimates. Students who were presented with a large number first returned a value of 2250. However, students who were presented with a low number first returned a value of 520. Anchors have robust effects even when they are highly irrelevant to the decision making task at hand (Yik et al., 2019). Furnham and Boo (2011) find that decision makers who encounter randomly generated numbers and incomplete computations are subject to the anchoring bias. For instance, numbers produced from a roulette wheel had a strong anchoring effect on individuals' estimates of the proportion of African countries in the United Nations (Tversky & Kahneman, 1974).

2.2. Anchoring bias in information systems

Unlike other disciplines that have long studied the anchoring bias, such as applied psychology, marketing, and medicine, research in this area is relatively nascent in information systems (IS) (Kordzadeh & Ghasemaghaei, 2021).

We reviewed the IS literature following the systematic review method.¹ The initial search produced a total of 219 articles out of 81,012 articles searched. We examined each article and identified 15 articles that tested the anchoring bias or anchoring effect (Table 1). Extant studies investigate the impact of the anchoring bias or effect in decision making in relation to algorithmic assessments (Bauer & Gill, 2023), software development effort estimation (Gren & Berntsson Svensson, 2021; Jørgensen & Halkjelsvik, 2010; Løhre & Jørgensen, 2016), software development performance (Mohan & Jain, 2008), requirements assessment in software development (Jayanth et al., 2011), crowd-funding (Kuo et al., 2021), product rating (Wang et al., 2021), cyber-attack management (Ceric & Holland, 2019), online shopping behavior (Wu et al., 2008; Wu & Cheng, 2011; Wu et al., 2012), consumers' constructed preference (Adomavicius et al., 2013), user competence (Marcolin et al., 2000), pricing (George et al., 2000), and collaborative consumption platforms (Bae & Koo, 2018).

¹ We began by searching the 55 IS journals with the rating of 4*, 4, 3, or 2 in the Academic Journal Guide 2021 list (journal list available upon request) using the search terms "anchoring bias," "anchoring and adjustment," "cognitive bias," "anchoring," "anchor," and "anchoring effect" in the fields of title, abstract, subject, or keywords within the period 1990–2023.

Table 1
Anchoring Effect in Relation to the Use of Information Systems.

Authors	Findings	Context	AI-related	Debias
(Bauer & Gill, 2023)	Disclosing the use of algorithmic scoring processes to individuals who are being scored affects their behaviors.	Algorithmic assessments	Yes	No
(Gren & Berntsson Svensson, 2021)	Obsolete requirements can anchor software implementation.	Software development effort estimation	No	No
(Kuo et al., 2021)	The 'bolstering range offer' pledge option design leads to the highest backing amount, while presenting the current average amount pledged can have a reversed anchoring effect on subsequent backers in physical goods crowdfunding projects.	Crowdfunding	No	No
(Wang et al., 2021)	The prior average rating of a product influences subsequent customers' ratings.	Product rating	No	No
(Ceric & Holland, 2019)	Cognitive biases limited the ABS's awareness of the eCensus system's vulnerabilities and hindered its preparation for and management of DDoS attacks.	Cyberattacks	No	No
(Bae & Koo, 2018)	Users tend to use low ratings and negative reviews as cues in information search.	Collaborative consumption platforms	No	No
(Løhre & Jørgensen, 2016)	This study finds no decrease in the anchoring effect with decreasing anchor precision or source credibility, suggesting that even implausible anchors from low-credibility sources can lead to anchoring effects.	Software development effort estimation	No	No
(Adomavicius et al., 2013)	The rating presented by a recommender system serves as an anchor for the consumer's constructed preference.	Consumer's constructed preference	Early DSS	No
(Wu et al., 2012)	This study finds the anchoring effect among people low in familiarity but not among people high in familiarity.	Online shopping	No	No
(Wu & Cheng, 2011)	Describing a product attribute in positive terms along with a high-	Online shopping	No	No

Table 1 (continued)

Authors	Findings	Context	AI-related	Debias
	anchor point induces more favorable responses than any other framing and anchoring combinations.			
(Jayanth et al., 2011)	A high-feedback effort can influence clients more toward a particular set of requirements in requirements assessment.	Requirements assessment in software development	No	No
(George et al., 2000)	Anchoring and adjustment biases remain strong within automated decision support systems, greatly impacting decision-making outcomes.	Pricing using DSS	Early DSS	Using warning
(Jørgensen & Halkjelsvik, 2010)	Software professionals provide significantly lower and more optimistic effort estimates when asked 'How much can be completed in Y work-hours?' compared to 'How much effort is required to complete X?'	Software development effort estimation	No	No
(Mohan & Jain, 2008)	Using traceability to mitigate cognitive biases in software development leads to higher quality solutions in software development tasks.	Software-development performance	No	Using traceability
(Wu et al., 2008)	The anchoring effect is robust; the repeated occurrence of an anchor does not reinforce the anchoring effect.	Online shopping	No	No
(Marcolin et al., 2000)	The frame of reference has an anchoring effect on participants' responses.	User competence	No	No

Notably, three studies investigate the impact of the anchor from early decision support systems (DSS) in the context of e-commerce. Adomavicius et al. (2013) find that the rating recommended by a DSS is an effective anchor impacting consumer preferences. George et al. (2000) find that a system-provided real estate listing price serves as a robust anchor for decision makers. More recently, investigating the impact of algorithmic assessments, Bauer and Gill (2023) uncover the nuanced relationship between algorithmic transparency and individual behaviors. This research suggests that disclosing algorithmic scores, especially when erroneous, can lead to self-fulfilling prophecies that drive individuals to act in accordance with those scores, illustrating that individuals can significantly base their actions and decisions on initial algorithmic assessments.

Two studies have explored strategies to mitigate the anchoring effect

in information systems contexts. Mohan and Jain (2008) examine the role of traceability in software development, finding that incorporating traceability features can reduce the impact of anchoring bias, thereby leading to more accurate and effective decision-making in complex software development tasks. George et al. (2000) investigate the effectiveness of warning mechanisms designed to alert users when their estimates closely align with system-generated recommendations. Their findings reveal that these warnings are insufficient to fully mitigate anchoring bias, as users continue to rely on the initial anchors despite the alerts.

Although existing research underscores the profound influence of the anchoring bias on decision making, there is limited empirical exploration of how anchoring and debiasing interventions function when the anchor originates from AI. Our research aims to explore how decision makers interact with AI-based anchors, particularly in the realm of human performance appraisal. Additionally, we explore the relatively uncharted domain of debiasing in the context of AI recommendations.

2.3. Anchoring effect on employee performance appraisal

Decision making in organizational management practices, such as employee performance appraisals, is frequently biased and error-prone (Battaglio Jr et al., 2019). Performance appraisals are an essential element of employee management through which organizations aim to optimize managerial practices and improve employees' behavior (Gill et al., 2019). Performance appraisals typically entail rating employee performance over time (Islami et al., 2018). However, supervisors are sometimes biased when rating employees and unintentionally taint the appraisal process. Several earlier experiments reveal the anchoring effects of human recommendations on employee performance appraisals (Battaglio Jr et al., 2019; Bellé et al., 2018; Schleicher et al., 2019). These experiments focus on testing variations in ratings of employees' performance when supervisors use different anchors (Bellé et al., 2017; Nagtegaal et al., 2020). The findings reveal that the anchoring bias is robust across various scenarios (Battaglio Jr et al., 2019). When using AI-assisted performance appraisal systems, supervisors might anchor their ratings to an AI-suggested score, such as a recommendation rating the employee 80 out of 100, which can unduly influence their final judgment to rate the employee closer to this anchor regardless of their independent assessment. Accordingly, we propose the following hypothesis:

Hypothesis 1. Participants in the high-anchor non-AI group will report significantly higher estimates than participants in the low-anchor non-AI group will.

In the past decade, there has been a surge in deploying AI solutions in organizational management, with machine-learning models assisting human experts in many high-stakes settings (Zhang et al., 2022). Organizations increasingly rely on AI to improve productivity, justice, and decisions (Wamba-Taguimdje et al., 2020). Human decision makers can use the recommendations of AI models to support managerial decisions.

An employee appraisal information system is a vital component of every organization, and an effective one improves employee satisfaction and productivity (Vrontis et al., 2022). Consequently, AI has been employed to manage employee performance appraisals, often providing data-driven recommendations to supervisors and other decision makers (Tambe et al., 2019). AI-assisted performance appraisal features communication between AI and humans (Leyer & Schneider, 2021). Recent studies highlight the need to understand and account for human cognition in AI-aided decision making. Kordzadeh and Ghasemaghaei (2021) stress that humans, as consumers of AI predictions, are the final decision makers in high-stakes applications. Human decisions can be distorted by AI output when individuals do not appropriately calibrate their dependence on AI (Green & Chen, 2019). Therefore, we propose:

Hypothesis 2. Participants in the high-anchor AI group will report

significantly higher estimates than participants in the low-anchor AI group will.

AI systems are increasingly recognized as effective tools for supporting organizational decision-making (Campbell et al., 2020). Decision makers posit that AI systems can logically, arithmetically, and consistently gather and process (e.g., weigh, prioritize, and consolidate) large amounts of data (Korteling et al., 2021). In performance appraisals, AI is specifically designed to mitigate human errors and biases that compromise the accuracy of evaluations. For instance, a supervisor with a positive relationship with an employee might unintentionally inflate the performance rating, resulting in an unjustifiably high score (Guo et al., 2021). In contrast, AI is less susceptible to emotional and stress-induced distortions that can affect human judgment (Wemm & Wulfert, 2017). Therefore, in high-anchor scenarios, when AI recommends a higher performance rating, decision makers are more likely to perceive this recommendation as objective and data-driven, free from the personal biases that often influence human judgment. Accordingly, anchoring effect occurs when decision makers rely heavily on the AI recommendation. The closer the final decision is to the anchor, the stronger the influence of the anchor. Therefore, we propose:

Hypothesis 3a. Participants in the high-anchor AI group will report estimates closer to the anchor than will participants in the high-anchor non-AI group.

The quality of leader-member exchange (LMX) relationships influences rater accountability (Park, 2017). Employees in low-quality LMX relationships tend to face more negative outcomes and interactions compared to those with high-quality LMX relationships (Rosen et al., 2011). In other words, supervisors in low-quality LMX relationships may unintentionally undervalue an employee's performance, unlike their evaluations of employees in high-quality LMX relationships. In contrast, AI tools are frequently perceived as more objective, processing performance data without the influence of personal biases (Wemm & Wulfert, 2017). This perceived neutrality of AI can lead decision makers to view AI-recommended low scores as more reliable than human-generated assessments. Consequently, supervisors are likely to align their ratings more closely with AI-generated low-anchors than with human-provided ones. Therefore, we propose:

Hypothesis 3b. Participants in the low-anchor AI group will report estimates closer to the anchor than will participants in the low-anchor non-AI group.

2.4. Debiasing with consider-the-opposite strategy

Biases in AI can lead to significant inequalities and discrimination, highlighting the urgent need for developing of effective and accountable mitigation strategies to ensure fair and equitable AI deployment (Isaksson, 2024). Debiasing is essential for improving decision-making by addressing the cognitive biases that often distort decision makers' judgments (Nagtegaal et al., 2020). Because cognitive biases work through different mechanisms, no single debiasing intervention works omnipotently on all types of biases (Cantarelli et al., 2020; Hilbert, 2012). Broadly, debiasing interventions can be classified into two main categories: 1) modifying the decision maker, which can be done by providing cues and education on a bias and its consequences (Lilienfeld et al., 2009), and 2) modifying the environment, thereby making the bias nonessential or mitigating its effect (Aleksavska et al., 2019).

Given the impact of the anchoring bias on decision making, several studies investigate interventions for debiasing (George et al., 2000; Mohan & Jain, 2008; Nagtegaal et al., 2020). Among the interventions, CTO is often preferred as a debiasing strategy mitigating anchoring bias because it effectively challenges the fixed cognitive reference frame created by anchoring bias (Aczel et al., 2015; Mumma & Wilson, 1995; Nagtegaal et al., 2020). Anchoring bias functions by creating a cognitive reference that can skew subsequent judgments, making it challenging

for decision makers to think beyond the initial anchor. CTO disrupts this frame by prompting individuals to consider why the anchor might be incorrect, thereby fostering more critical thinking and opening the path to alternative perspectives (Aczel et al., 2015). This approach has proven effective across various domains, including election estimates (Mussweiler et al., 2000), education (Adame, 2016), and performance appraisals (Nagtegaal et al., 2020).

Additionally, the simplicity and low cost of implementing CTO make it a practical and attractive method for reducing bias. By encouraging decision makers to list reasons why an anchor might be inappropriate, CTO can lead to more balanced and thoughtful decision-making processes (Adame, 2016). This strategy is particularly advantageous in contexts involving AI-generated anchors, where managers might otherwise heavily rely on initial AI recommendations. CTO helps managers look beyond AI suggestions, promoting fairer and more accurate evaluations (Van Brussel et al., 2020). In high-anchor scenarios, where AI-generated recommendations suggest higher performance ratings, decision makers who initially rely too heavily on the AI's recommendation—resulting in inflated evaluations—are prompted by the CTO strategy to critically reassess these recommendations. The strategy helps them identify potential overestimations or errors, leading to a more balanced and accurate evaluation (Nagtegaal et al., 2020). Therefore, we propose:

Hypothesis 4a. . Participants in the CTO high-anchor AI group will report estimates that are lower than estimates from participants in the non-CTO high-anchor AI group.

AI-generated recommendations may occasionally suggest low performance ratings. In these low-anchor scenarios, the CTO strategy encourages managers to critically assess the validity of these low scores, prompting them to look beyond the initial recommendation (Nagtegaal et al., 2020). This approach is particularly valuable when assessing employees with short work histories or when the AI recommendation is derived from incomplete data (IBM, 2023). In such instances, the AI-generated rating might not fully capture the employee's full contributions, potentially leading to an undervaluation of the employee's performance. By implementing the CTO strategy, decision makers engage in a more thoughtful evaluation process, reducing the likelihood that the low-anchor will unduly influence the final decision (Adame, 2016; Nagtegaal et al., 2020). Therefore, we propose:

Hypothesis 4b. . Participants in the CTO low-anchor AI will report estimates that are higher than estimates from participants in the non-CTO low-anchor AI group.

3. Research design

This study employs a mixed-method approach to explore the effect of anchoring bias in AI-assisted decision making and to study how to debias the effect. Our research design consists of three interconnected studies, each addressing specific objectives that collectively contribute to a comprehensive understanding of the research problem.

Study 1 aims to test the presence and strength of the anchoring effect of AI recommendations. We employed a 2 (Recommendation Source: AI vs. Human) $\times$ 2 (Anchor: High vs. Low) factorial experimental design. Participants were asked to evaluate an employee's performance based on scenarios with different anchors and recommendation sources. The study quantitatively examines the impact of AI-based performance appraisal recommendations on decision making through the experiment.

Study 2, building on the findings of Study 1, introduces a debiasing intervention, specifically the CTO strategy, to mitigate the anchoring bias in relation to AI recommendations. This study also uses a 2 $\times$ 2 factorial design to assess the effectiveness of the CTO strategy in both high and low-anchor scenarios. Participants were exposed to AI-generated anchors with and without the CTO intervention to evaluate

whether this strategy reduces the anchoring bias in performance appraisals.

Study 3 aims to validate the experimental findings from the previous studies, exploring how AI recommendations influence decision making in practice and assessing the effectiveness of the CTO strategy in real-world scenarios. This study adopts a qualitative approach using the Critical Incident Technique (CIT) to gather in-depth insights from managers with real-world experience in AI-assisted decision making.

By integrating experimental and qualitative data, this mixed-method approach allows us to triangulate our findings, enhancing the generalizability of the results (Venkatesh et al., 2023). This comprehensive view of how anchoring bias operates in AI-assisted decision making and how it can be effectively mitigated provides valuable insights.

4. Study 1: anchoring and adjustment bias

4.1. Overview

Study 1 explored how an AI-based performance appraisal recommendation serves as an anchor that impacts decision making. Specifically, it (1) investigated the impact of an AI-based recommendation on performance appraisal and (2) tested whether participants behave differently with high- versus low-anchors in an AI versus non-AI scenarios.

Given the complexity of human cognition in decision making, a controlled experiment, which allows for precise control of extraneous variables (Mitchell & Jolley, 2013), can reduce the impact of confounding factors and provide high internal validity (Mangalaraj et al., 2014; Nagtegaal et al., 2020). We utilized a 2 (AI/Human Recommendation) $\times$ 2 (High /Low-Anchor) factorial experimental design (see Fig. 1).

4.2. Method

We tested the anchoring effect in the setting where a supervisor rates the performance of a subordinate. We adapted the experiment design, measures, and procedure from Bellé et al. (2017) because their robust experimental framework has proven effective in exploring cognitive biases in performance appraisal contexts, making it highly suitable for extending into the AI context. We extended their study by adding the AI scenarios. Participants were asked to imagine themselves as the supervisor of the employee depicted in the vignette.

Using four scenarios, we varied the recommendation source (e.g., human vs. AI) and anchor (e.g., high vs. low) assigned to the subordinate (see Appendix A). Participants were randomly assigned to receive either a high-anchor or a low-anchor embedded within the scenario. Specifically, the high-anchor group received a recommendation of '91 out of 100' as the previous year's performance rating, while the low-anchor group received '51 out of 100'. The dependent variable was the performance appraisal score on a 0–100 scale. We designed the scenarios in

		Anchor	
		High	Low
Recommendation Source	Human	high-anchor, non-AI group N = 93	low-anchor, non-AI group N = 100
	AI	high-anchor, AI group N = 90	low-anchor, AI group N = 94

Fig. 1. Two-by-Two Factorial Design for Study 1.

collaboration with a senior scholar experienced in supervision, reviewing the narrative and anchors to confirm their alignment with organizational practices.

Participants were recruited using Prolific, a reputable crowdsourcing platform tailored for academic research (Alorwu et al., 2021; Gordon, 2021; Palan & Schitter, 2018; Peer et al., 2017); Prolific provides access to a diverse general population, facilitating large-scale online recruitment (Fink, 2022).

In April 2022, we used the pre-screening option on Prolific to recruit people in the United States who had supervisory experience. This resulted in 423 participants, of which 377 provided valid responses after passing quality checks (e.g., completeness check, manipulation check, attention check, and Prolific ID check). Of the respondents providing valid responses, 50.1 % were female, 48.0 % were male, 1.6 % were nonbinary, and 0.3 % preferred not to reveal. These respondents included 30.7 % with a high school or associate's degree, 44.6 % with a bachelor's degree, and 24.7 % with a graduate degree or higher. 78.2 % were in supervisory roles at the time of the survey, while 21.8 % had such experience in previous positions. The ages ranged broadly from 18 to over 65, with 68.4 % being between 25 and 44 years old. Detailed demographics are provided in Table 2.

4.3. Results

First, the high- and low-anchor non-AI groups (replication groups) were compared with a *t*-test (testing Hypothesis 1). Testing Hypothesis 1 to replicate earlier findings is essential for extending the generalizability of results, as outcomes from one population or context may not necessarily apply to another (Nagtegaal et al., 2020). The replication is successful if a significant difference is obtained in the same direction as prior studies (Cantarelli et al., 2020). Second, the high- and low-anchor AI groups were compared with a *t*-test (testing Hypothesis 2). The AI recommendations' anchoring effect exists if the mean of the low-anchor AI group is significantly lower than the high-anchor AI group. Third, we compared the AI groups and the non-AI groups in the high-anchor and low-anchor scenarios, respectively.

We calculated effect sizes, guided by Lakens (2013), and report the magnitude of the effect sizes following the rules of thumb offered by Sawilowsky (2009). Sawilowsky (2009) expanded Cohen's criteria for reporting effect sizes by very small (0.01), small (0.2), medium (0.5), large (0.8), very large (1.2), and extremely large effects (2.0).

The results support Hypothesis 1: Participants in the high-anchor non-AI group report significantly different estimates from those in the low-anchor non-AI group. In other words, the replication was successful: the mean rating for the high-anchor condition ($M = 92.44$, $SD = 3.208$, $N = 93$) is significantly higher ($t = 19.80$, $p < .001$) than the rating for

the low-anchor condition ($M = 71.27$, $SD = 9.836$, $N = 100$). The means of the non-AI groups are shown in Table 3. The effect size is extremely large (Cohen's $d = 2.85$).

The results also support Hypothesis 2, showing that participants in the high-anchor AI group report estimates that are significantly different from participants in the low-anchor AI group. This suggests that the AI recommendations' anchoring effect exists. The mean rating for the high-anchor condition ($M = 89.64$, $SD = 4.786$, $N = 90$) is significantly higher ($t = 14.84$, $p < .001$) than the rating for the low-anchor condition ($M = 72.66$, $SD = 9.794$, $N = 94$). The effect size is extremely large (Cohen's $d = 2.19$).

We further tested whether there is a significant difference in performance appraisal with an AI recommendation versus a non-AI recommendation (Hypothesis 3a for a high-anchor and Hypothesis 3b for a low-anchor). The results support Hypothesis 3a; participants in the high-anchor AI group reported estimates closer to the anchor than did participants in the high-anchor non-AI group. The estimates produced in the high-anchor AI group ($M = 89.64$, $SD = 4.786$, $N = 90$) are significantly ($t = 4.66$, $p < .001$) closer to the anchor than in the high-anchor non-AI group ($M = 92.44$, $SD = 3.208$, $N = 93$), with a moderate effect size (Cohen's $d = 0.689$) (Appendix B). In addition, we tested Hypothesis 3b. The estimates produced in the low-anchor AI group ($M = 72.66$, $SD = 9.794$, $N = 94$) are not significantly different ($t = 0.99$, $p = .163$) from those produced in the low-anchor non-AI group ($M = 71.27$, $SD = 9.836$, $N = 100$); thus, H3b was not supported. The findings show there is a significant difference between the AI recommendation and non-AI recommendation in the high-anchor scenario but not in the low-anchor scenario. In other words, the differences in AI and non-AI anchoring effects are circumstance dependent.

Fig. 2 indicates the mean of performance appraisal scores is lower for the AI recommendation than non-AI recommendation when the recommended score is high. Conversely, the mean of performance-appraisal scores is higher for AI recommendations than non-AI recommendations when the recommended score is low. In other words, the recommendation source interacts with the anchor to impact decision makers.

5. Study 2: CTO debiasing strategy

5.1. Overview

Study 1 revealed AI recommendations significantly impact decision making in performance appraisals, particularly highlighting a stronger anchoring effect in high-anchor scenarios compared to non-AI recommendations. Given the presence of an AI anchoring bias, we conducted Experiment 2 to explore an approach to mitigate that bias. Specifically, Experiment 2 aimed (1) to evaluate the CTO strategy's effectiveness in mitigating the anchoring bias associated with AI recommendations, and (2) to determine how the CTO strategy performs across different anchor scenarios (high- versus low-anchors). This study employed a 2 (High/Low-Anchor) $\times$ 2 (With/Without CTO) factorial design (Fig. 3).

5.2. Method

The experiment was conducted within the context of performance

Table 2
Subject Demographic Characteristics of Study 1.

Demographic Variables	Category	Count	Percentage
Gender	Female	189	50.1 %
	Male	181	48.0 %
	Non-binary	6	1.6 %
	Prefer not to say	1	0.3 %
Education	High school or less	59	15.6 %
	Associates or technical degree	57	15.1 %
	Bachelor's degree	168	44.6 %
	Graduate or higher	93	24.7 %
Supervision experience	In current position	295	78.2 %
	In prior positions	82	21.8 %
Age	18–24 years old	37	9.8 %
	25–34 years old	134	35.5 %
	35–44 years old	124	32.9 %
	45–54 years old	52	13.8 %
	55–64 years old	25	6.6 %
	65 years and over	5	1.3 %

Table 3
Descriptive Statistics of Study 1.

Group	N	Mean	Minimum	Maximum	Standard Deviation
High-anchor, non-AI	93	92.44	80	98	3.208
Low-anchor, non-AI	100	71.27	51	95	9.836
High-anchor, AI	90	89.64	70	98	4.786
Low-anchor, AI	94	72.66	50	90	9.794

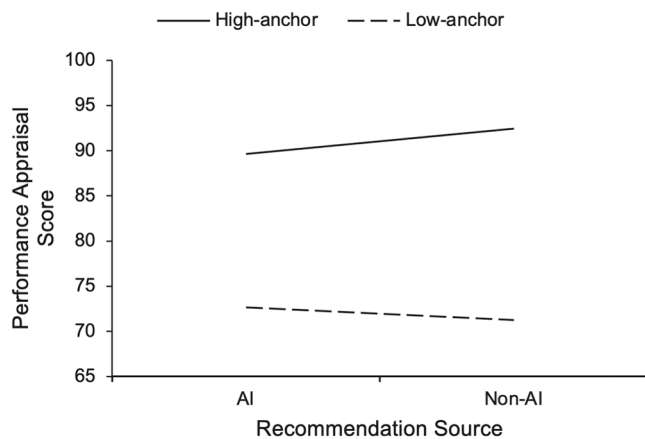


Fig. 2. Estimates from AI Recommendation and Non-AI Recommendation.

		Anchor	
		High	Low
Intervention	Without CTO	non-CTO, high-anchor group N = 99	non-CTO, low-anchor group N = 98
	With CTO	CTO, high-anchor group N = 100	CTO, low-anchor group N = 101

Fig. 3. Two-by-Two Factorial Design for Experiment 2. Note: all four groups are using AI recommendations.

appraisals, where participants assumed the role of a supervisor. We adapted the debiasing intervention from earlier CTO experiments (Adame, 2016; Nagtegaal et al., 2020) because the CTO strategy has demonstrated robust efficacy in reducing anchoring bias, especially in managerial decision-making contexts similar to those in our study. In line with Nagtegaal et al. (2020), respondents were asked to list two reasons the anchor may be inappropriate. The direction in which the anchor was inappropriate was indicated. In other words, if the anchor was too low, participants were asked to consider why it may be too low. If the anchor was too high, participants were asked to explain why it may be too high.

The experiment had four scenarios: non-CTO high-anchor, non-CTO low-anchor, CTO high-anchor, and CTO low-anchor. All participants received AI recommendations as part of the task, but the presence of the CTO intervention varied between scenarios (Appendix C). The dependent variable was the performance score on a 0–100 scale.

We used the pre-screening option on Prolific to select people in the United States who have engaged in supervisory duties. The operational procedure was the same as in Study 1. This resulted in 439 participants, with 398 providing valid responses after quality checks. Of the respondents providing valid responses, 48.2 % were female, 50.0 % were male, and 0.5 % identified as non-binary, with 1.3 % preferring not to disclose. Among these 398 respondents, 29.6 % had a high school or associate's degree, 43.5 % held a bachelor's degree, and 26.9 % had a graduate degree or higher. At the time of the survey, 79.4 % were in supervisory roles, while 20.6 % had supervisory experience in previous positions. The age distribution ranged from 18 to over 65 years, with 61.8 % of participants between 25 and 44 years old. Detailed demographics are provided in Table 4.

Table 4

Subject Demographic Characteristics of Study 2.

Demographic Variables	Category	Count	Percentage
Gender	Female	192	48.2 %
	Male	199	50.0 %
	Non-binary	2	0.5 %
	Prefer not to say	5	1.3 %
Education	High school or less	59	14.8 %
	Associates or technical degree	59	14.8 %
	Bachelor's degree	173	43.5 %
	Graduate or higher	107	26.9 %
Supervision experience	In current position	316	79.4 %
	In prior positions	82	20.6 %
Age	18–24 years old	36	9.1 %
	25–34 years old	140	35.2 %
	35–44 years old	106	26.6 %
	45–54 years old	71	17.8 %
	55–64 years old	35	8.8 %
	65 years and over	10	2.5 %

5.3. Results

The experiment had two steps. First, the replication groups (i.e., non-CTO high- and low-anchor groups) were compared with a *t*-test. The replication is successful if a significant difference is obtained in the same direction as in Experiment 1. Consistent with Experiment 1, we calculated effect sizes, as guided by Lakens (2013), and report the magnitude of the effect sizes the rules of thumb from Sawilowsky (2009). The results reveal our replication in Experiment 2 is successful, as there is a significant difference between the low-anchor AI group and the high-anchor AI group ($t = 16.22, p < .001$). The effect size is extremely large (Cohen's $d = 2.31$).

Second, using *t*-tests, we compared the replication groups with the corresponding CTO high- and low-anchor debiasing groups. Debiasing is effective if the CTO high-anchor AI group reports significantly lower means than the high-anchor AI group without the CTO intervention (Hypothesis 4a) and if the mean of the CTO low-anchor AI group is significantly higher than the low-anchor AI group without the CTO intervention (Hypothesis 4b).

Hypothesis 4a. is supported: the CTO intervention generates estimates ($M = 84.64, SD = 7.201, N = 100$) that are significantly ($t = 7.31, p < .001$) lower than the high-anchor group without the intervention ($M = 90.58, SD = 3.665, N = 99$), as shown in Table 5. The effect size is large (Cohen's $d = 1.04$). However, Hypothesis 4b is not supported: the estimates produced in the CTO low-anchor AI group ($M = 75.58, SD = 9.076, N = 101$) are not significantly different ($t = 2.11, p = .982$) from those produced in the low-anchor group without the CTO intervention ($M = 72.67, SD = 10.345, N = 98$) (Appendix D). The results suggest that CTO interventions effectively debias performance ratings based on AI recommendations by significantly adjusting the estimates in high-anchor scenarios.

The interaction plot (Fig. 4) reveals an interaction effect between the CTO intervention and the anchor (high vs. low). In other words, the estimates are lower with CTO than without CTO when the anchor is

Table 5

Descriptive Statistics of Study 2.

Group	N	Mean	Minimum	Maximum	Standard Deviation
Non-CTO, high-anchor	99	90.58	75	98	3.665
Non-CTO, low-anchor	98	72.67	51	95	10.345
CTO, high-anchor	100	84.64	55	100	7.201
CTO, low-anchor	101	75.58	50	100	9.076

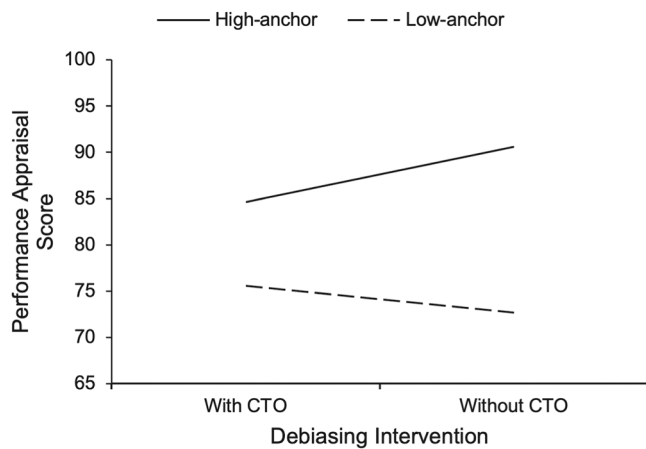


Fig. 4. Estimates with CTO and without CTO.

high. The intervention interacts with the anchor to impact decision makers. Table 6 summarizes the results of both studies.

6. Study 3: critical incident technique

6.1. Overview

Study 3 aims to validate prior results using the CIT, exploring the impact of AI recommendations on decision making. Additionally, it investigates whether AI-based anchoring bias can be mitigated through the CTO strategy. CIT involves prompting participants to recall and describe a time when a specific incident, action, or behavior influenced a particular outcome (Kari et al., 2020; Thomas & Bostrom, 2010). CIT can capture detailed incident descriptions for profound insights, making it suitable for examining in depth how AI affects managerial decision making processes and validating our results from previous studies (Mulcahy et al., 2024).

6.2. Method

To collect data, we conducted an online survey of people with managerial experience. Participants were asked to reflect on specific incidents where they used AI for decision making, focusing on instances where an initial numerical estimate provided by AI impacted their decision. For instance, one prompt asked participants to “Think of a time when you had a notably positive or negative experience with using AI to

assist in decision making, specifically when you were provided with an initial numerical estimate. Reflect on an instance where this initial numerical estimate impacted your decision making process.” The questions explored the AI tool they used, the nature of their decision, the context and outcome of the decision, and their perceptions of the role of AI. Additionally, participants were asked about their experiences with disregarding or considering the opposite of AI recommendations and how these strategies affected their final decisions. In addition to the CIT questions, participants responded to questions related to their general perceptions of AI using the Artificial Intelligence Scale (GAAIS) (Schepman & Rodway, 2023). These questions assessed their overall beliefs in AI and their views on the reliability and effectiveness of AI in decision making.

Of the recruited 45 participants, responses were checked for completeness. Incomplete responses and those that only talked about AI in general without reflecting on a specific use of a numeric anchor were removed, resulting in 24 valid responses. The respondents had an average age of 35.17 (SD = 12.87), with 41.7 % identifying as female and 58.3 % as male. A majority of 70.8 % resided in the United Kingdom, followed by 16.7 % in the United States, and 4.2 % in Australia. The GAAIS score, which assessed their overall beliefs in AI and their views on the reliability and effectiveness of AI in decision making, had a mean of 3.40 (SD = 0.36).

The research team systematically coded participants’ open-text responses. This coding process focused on key aspects of their experiences with decision making using AI recommendations. Specifically, we coded whether participants relied on the AI anchor, whether the AI anchor impacted their decision, the reasons for relying on AI, whether they had experience with the CTO strategy, and whether the CTO strategy impacted their decision making. After coding the responses, we identified patterns that illustrate how AI recommendations influence decision making processes and how effectively the CTO strategy mitigates anchoring biases. This comprehensive coding and analysis following CIT guidelines validates the experimental results from the earlier studies.

6.3. Results

The analysis reveals that 23 out of 24 participants (95.8 %) reported that they had experience relying on AI recommendations to make decisions, emphasizing the impact of these recommendations on their final judgments. For example, one participant noted, “I used an AI tool to estimate the potential revenue increase from launching a new product line. The AI predicted a 20 % increase in revenue, which significantly influenced our decision to proceed with the launch. This anchor helped us set realistic expectations and allocate resources more effectively.” Such reliance on AI anchors underscores the significance of AI in shaping decision making processes and highlights the need for accurate and reliable AI-generated anchors to ensure sound decisions.

Among the 23 participants who have experience relying on AI anchors, 21 (91.3 %) believed that the impact of the AI recommendations was positive. Conversely, 2 participants (8.7 %) reported negative experiences when AI recommendations were impractical. For instance, one participant recounted, “I was using an AI tool to help estimate the cost of a home renovation project. The AI provided an initial numerical estimate of \$15,000 for the entire renovation based on the details I inputted about the size of the house, the materials needed, and the scope of the work. However, as the project progressed, the actual costs significantly exceeded the AI’s estimate, ultimately totaling \$25,000.” This highlights the potential limitations of AI in complex decision making scenarios, emphasizing the need for human oversight and additional verification to ensure accurate outcomes.

The data also revealed the potential of the CTO strategy in mitigating AI-based anchoring bias. Out of the 23 participants using AI anchors, 11 (47.8 %) had some experience using the CTO techniques for cautious decision making. 10 of these 11 participants (90.9 %) found the CTO techniques useful, while only one believed that AI had already done a

Table 6
Hypotheses Test Results.

Item	Hypothesis	Result
H1	Participants in the high-anchor non-AI group will report estimates that are significantly higher than estimates from participants in the low-anchor non-AI group.	Supported * **
H2	Participants in the high-anchor AI group will report estimates that are significantly higher than estimates from participants in the low-anchor AI group.	Supported * **
H3a	Participants in the high-anchor AI group will report estimates that are closer to the anchor than estimates from participants in the low-anchor non-AI group.	Supported * **
H3b	Participants in the low-anchor AI group will report estimates that are closer to the anchor than estimates from participants in the low-anchor non-AI group.	Not Supported
H4a	Participants in the CTO high-anchor AI group will report estimates that are lower than estimates from participants in the non-CTO high-anchor AI group.	Supported * **
H4b	Participants in the CTO low-anchor AI group will report estimates that are higher than estimates from participants in the non-CTO low-anchor AI group.	Not Supported

Notes: * ** $p < 0.001$; * $p < 0.05$; * $p < 0.01$

good job. One participant described an instance where considering the opposite of an AI recommendation influenced their decision making: “I recall a situation where an AI system recommended investing in a certain stock based on historical trends, but I decided to consider the opposite due to conflicting market analysis from human experts. Despite the AI’s recommendation, I opted to diversify my portfolio differently, relying on a blend of AI insights and human judgment. This approach led to a more balanced investment strategy, reducing risk exposure and yielding competitive returns compared to instances when I solely relied on either AI or human expertise.”

Noticeably, a participant highlighted the importance of adjusting AI recommendations for improved performance: “Once the AI recommended reducing marketing spend, but I chose to increase it based on market conditions. Our campaign was a success and boosted brand awareness. I think it’s important to balance AI insights with human judgment rather than solely relying on AI recommendations.” This underscores the importance of trust and user confidence in AI systems, suggesting that AI tools must demonstrate consistent accuracy and reliability to be fully embraced in decision making processes.

In summary, the qualitative analysis through CIT provided rich insights into how users perceive AI recommendations, validating the results of Studies 1 and 2. By triangulating these findings with quantitative data, we gained a comprehensive understanding of the factors that enhance or undermine the perceived reliability and usefulness of AI in decision making. This study also highlighted the potential of the CTO strategy to mitigate AI-based anchoring bias, demonstrating its effectiveness in promoting more balanced and informed decision making.

7. Discussion

Our study explored the AI anchoring bias in performance appraisals, revealing how AI-generated recommendations can significantly influence supervisors’ ratings of subordinates. The results demonstrated that supervisors exposed to high AI-recommended scores rated employee performance substantially higher compared to those exposed to low AI-recommended scores. This effect was more pronounced with AI-generated anchors than with human-generated anchors, suggesting that AI’s perceived objectivity or reliability might amplify its anchoring influence. Specifically, when given a high-anchor, supervisors’ ratings in the AI scenarios were significantly closer to the provided anchor than those in the human scenarios. Conversely, when exposed to low-anchors, the difference in ratings between the AI and non-AI scenarios was not significant. One plausible reason for this circumstance-dependent effect could be that supervisors conducting performance appraisal perceive AI recommendations as more accurate or reliable where high-anchors are provided. Notably, this subgroup analysis should be interpreted with caution, as the contextual factors are more complex in real-world scenarios. This asymmetry underscores the complexity of AI’s impact on decision-making processes, as it can elevate performance appraisals with high-anchors without equivalently depressing them with low-anchors.

In addition, our study evaluated the efficacy of the CTO strategy in mitigating AI-generated anchoring bias. The results revealed that the CTO intervention effectively reduced the anchoring bias for high AI-generated recommendations. Supervisors using the CTO strategy provided significantly lower ratings than those who did not. This finding highlights the potential of the CTO strategy to challenge the initial AI anchor and encourage more critical evaluation in performance appraisal. However, the CTO strategy did not significantly affect ratings in the low-anchor scenarios. A plausible reason could be that when supervisors are presented with low-anchors, they may naturally engage in critical thinking or cautious consideration before finalizing their judgments. This pre-existing conservative behavior can reduce the apparent impact of debiasing interventions like the CTO strategy. These findings illustrate the potential for targeted debiasing strategies to enhance decision-making accuracy when using AI in performance appraisals.

Future research should further explore how decision makers’ perceptions of AI objectivity impact their engagement with debiasing strategies like CTO, particularly when dealing with low-anchors.

Together, these experiments underscored the need for organizations to carefully consider the design and implementation of AI systems in performance appraisal. Our findings were further supported by a qualitative study employing the critical incident technique, which validated the experimental results by offering real-world insights into the impact of AI on decision-making and demonstrating the practical effectiveness of the CTO strategy. Our study provides a foundation for further research into the intersection of AI, cognitive bias, and organizational decision-making, emphasizing the importance of developing and integrating robust debiasing mechanisms into AI-based decision support systems.

7.1. Implications for research

Our work offers the first experimental test of the presence and salience of the anchoring effect of AI recommendations on employee performance appraisals. Joining recent efforts to explore cognitive biases in managerial operations, we substantially broaden the IS community’s understanding of AI recommendations in organizational decision making. Complementing existing evidence of anchoring biases in performance appraisal, our results suggest that considering the opposite can help debias AI recommendations.

Firstly, as one step further toward understanding human cognition in relation to AI recommendations for employee performance appraisal, our study explores how AI recommendations influence supervisors’ decision-making. It identifies a noticeable pattern of individuals’ judgmental deviation from rationality based on their perception of the AI recommendations. In other words, AI recommendations may dictate their behavior in organizational decision making, resulting in distorted and inaccurate judgment. While anchoring biases and their effects on decision making are widely studied, our study notes that AI-assisted decision making introduces a new paradigm, in which it is important to systematically examine the role of human cognitive biases. In line with antecedent experiments in the non-AI scenarios, our study does not conclude that we would observe the same anchoring pattern if we employed other treatments (e.g., qualitative anchors) and/or used different measures to rate employees’ performance (e.g., visualized performance matrix). Therefore, future research could explore alternative design parameters, such as varying the context, score, or performance year, to generate deeper insights into how AI recommendations impact decision-making biases across diverse scenarios. Additionally, exploring how individual differences, such as technology readiness, concerns about algorithms, and trust in AI, as well as underlying psychological mechanisms, impact the susceptibility to anchoring biases could provide deeper insights into personal factors that moderate these effects.

Secondly, our study contributes to the literature on the CTO strategy by testing its effectiveness in mitigating the bias of AI-generated high-anchors. We provide new empirical evidence on the utility of CTO beyond its traditional applications. Specifically, our findings demonstrate that the CTO strategy effectively reduces the influence of high AI-generated anchors, which often skew decision-making in performance appraisals. This suggests that CTO is not only effective in generic decision-making scenarios but also holds promise in the increasingly prevalent domain of AI-assisted performance appraisals. Future research can build upon these insights by examining how the CTO strategy can be adapted to mitigate other types of biases introduced by AI and by exploring its applicability in different organizational and technological contexts where AI recommendations are used.

Thirdly, this paper establishes a multidisciplinary foundation for theoretical research at the intersection of cognitive bias, AI, and organizational decision making. By establishing the significance of the anchoring effects of AI recommendations, we present a foundation upon which future research can build. For example, future research could

investigate how AI recommendations bias various stakeholders, including executives, managers, data analysts, system designers, and other employees. Moreover, examining the interplay between traditional IS constructs, such as perceived risks and trust in AI, with newer concepts like AI heuristics, cognitive biases, and ethical decision-making, can provide valuable insights. Additionally, AI recommendations are also widely adopted in judicial decision making and in other scenarios. It would also be interesting to test the anchoring effects of AI recommendations in relation to these different use cases.

7.2. Implications for practice

The use of AI in sensitive areas, including hiring, healthcare, and criminal justice, has grown, raising debates about bias and fairness in AI-assisted decision making. As human decision making is often unconsciously shaped by individual and societal biases, optimists believe decision making aided by AI can be less biased. Our study finds that AI recommendations' anchoring effect is salient in decision making concerning performance appraisal, even when minimal information is given to describe AI performance and trustworthiness (e.g., accuracy and user satisfaction). This is particularly concerning when the AI recommendations are based on flawed data, for example, the AI is trained based on human decisions that may already be biased.

In this sense, AI recommendations, as anchors, hold great potential for amplifying bias rather than reducing it, especially when the anchors are low quality and embed human and societal biases (e.g., recommendations from a hiring algorithm based on analyzing previous human decisions that penalized applicants in a minority). Therefore, both AI system designers and decision makers relying on AI innovations should be aware of such salient AI anchoring effects. When designing or consuming AI solutions, stakeholders should be mindful of errors and biases in relation to AI. In the performance management context, biases could lead to poor performance reviews that ultimately impact an employee's pay or job stability.

Accordingly, AI systems should be viewed as tools that require careful oversight, especially in critical business operations such as employee performance appraisals. The presence of anchoring bias in AI-assisted decisions highlights the need for a thoughtful integration of human judgment alongside AI recommendations. To address this, decision-making environments should encourage reflective thinking and incorporate debiasing strategies. Organizations can achieve this by implementing regular training sessions that raise awareness of cognitive biases, such as anchoring effects, and how these biases can emerge in AI-assisted decision-making. Providing decision makers with the knowledge and tools to recognize and mitigate bias will promote more equitable and data-driven outcomes.

Decision makers should recognize that incorporating effective debiasing interventions is a crucial step toward reducing the adverse impacts of AI-related cognitive biases in their organizations. In particular, managers can implement appropriate strategies to critically assess AI-generated outputs, ensuring they do not rely solely on these recommendations without considering alternative viewpoints and additional data sources. For instance, decision makers can apply the CTO strategy, considering reasons why an AI-generated anchor might be inaccurate, to reduce bias in performance appraisals.

Moreover, when designing an AI-based decision support system, the designer can take into account the biases and limitations inherent in the anchor-based decision making process. AI systems should also include built-in functions and interfaces that identify and eliminate biases in underlying data and algorithms, thereby providing as accurate and fair recommendations—and anchors—as possible to human decision makers, and thus promoting fairer decisions. For example, AI systems could be designed to periodically audit their recommendations, flagging potential biases in the output. This could be achieved through machine learning models that continually learn from feedback and adjust their algorithms to reduce biased outcomes over time. Additionally, AI

solution designers can provide cues or heuristics in system dashboards to mitigate potential AI anchoring bias from low-quality anchors. For example, AI systems may include cautions about algorithmic performance (e.g., precision, recall, accuracy, sensitivity, and ROC), especially when the performance is moderate or less than moderate. AI systems may also send alerts to users before decision makers finalize impetuous decisions based on AI recommendations. CTO interventions can also be embedded into the system. For instance, an AI-aided performance appraisal system could include features that prompt supervisors (e.g., using dialog box) to consider alternative perspectives before submitting their final evaluations. In particular, the system could guide decision makers to systematically and structurally consider the opposite of the AI-recommended results, encouraging more critical reflection and ultimately leading to fairer outcomes in employee evaluations.

7.3. Limitations and future research

Our research, inevitably, has limitations. Its main limitation is that, like other controlled experiments with high internal validity, it was conducted using simplified scenarios and an online sample. Real-world scenarios might be more complex, which could lead to more intricate phenomena in relation to AI and the CTO strategy. Accordingly, we call for future research to replicate our study in more natural settings and other contexts. Additionally, we encourage scholars to test our experimental manipulations with different samples of managers across varied contexts.

Another limitation of our research could be the use of the term "bias." "Bias" implies a relationship between cognitive heuristics and adverse outcomes. [Gigerenzer \(1991\)](#) notes that cognitive biases can be useful decision making tools. In this study, we use the term "bias" consistent with the traditional discussion in [Kahneman \(2011\)](#) and [Tversky and Kahneman \(1974\)](#), as well as with antecedent studies on the anchoring bias, agreeing that heuristics can lead to suboptimal judgments as indicated by expected utility theory.

Moreover, our study primarily examined AI-generated versus self-generated recommendations. We did not explore the differences between self-generated recommendations and those provided by others, such as peers. Investigating these distinctions could provide more nuanced insights. Therefore, we call for future research to further explore the differences between AI and different sources of human recommendations.

Finally, we only investigate anchoring bias. IS scholars can extend our work by testing how AI is associated with diagnostic inaccuracies or suboptimal decisions from other cognitive biases, such as the availability bias, confirmation bias, overconfidence, framing, and many other biases. In addition, we highlight opportunities to investigate how human cognitive bias is related to AI algorithms and data, as well as to the many AI design choices (e.g., level of transparency, interpretability, and how dashboards present the results). It would also be interesting to study the synergizing effect of a combination of these biases in AI-aided organizational decision making. Future research can also enrich IS theories and models by exploring cognitive biases of AI-aided decision makers with different personality traits.

While we tested the effect size of the anchoring bias in relation to AI, we did not find or define any unified scales measuring the bias. Therefore, we note that the results of this study are to be interpreted with caution. Given that our findings are only tested with controlled experiments in performance appraisal, we call for future research to explore the metrics that can measure the impact of the AI-related cognitive bias so that these biases can be compared across a variety of organizational fields.

8. Conclusions

Our study investigated whether the AI anchoring bias appears in decisions in the institutional context, whether there are differences

between the AI and non-AI recommendations, and whether supervisors can be debiased by a low-cost, low-intensity CTO intervention. Organizations should recognize the significance of decision makers' cognitive biases and utilize debiasing methods, such as CTO, to mitigate biases in managerial operations.

Our research serves as a springboard to establish IS and psychology theories at the intersection of AI, cognitive bias, and organizational justice. We encourage IS researchers to further explore the phenomena of cognitive biases in the adoption of information systems, especially AI. A stream of systematic research on AI-associated cognitive biases would provide new insights that may affect industries and help attenuate managerial errors in organizations. IS scholars should work across disciplines to develop a comprehensive understanding of cognitive heuristics and other psychological factors in relation to the use of AI

recommendations; an interdisciplinary view can help system designers produce better AI solutions, thereby boosting societal and organization fairness through both long-term and short-term AI use.

CRediT authorship contribution statement

Carter Lemuria: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Liu Dapeng:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

None.

Appendix A. Experiment 1 Adapted from Bellé et al. (2017)

Non-AI Recommendation Scenarios:

Imagine that you have to assess this year's performance of a subordinate of yours. During this year, your subordinate met the majority of goals, had good interpersonal skills with his colleagues, and showed moderate creativity in proposing new ideas for the improvement of the services.

Your company does a performance review every year. The previous year, you assigned your subordinate a performance rating of 51/100[91/100]. You have to decide whether to assign a rating lower or higher than 51/100[91/100].

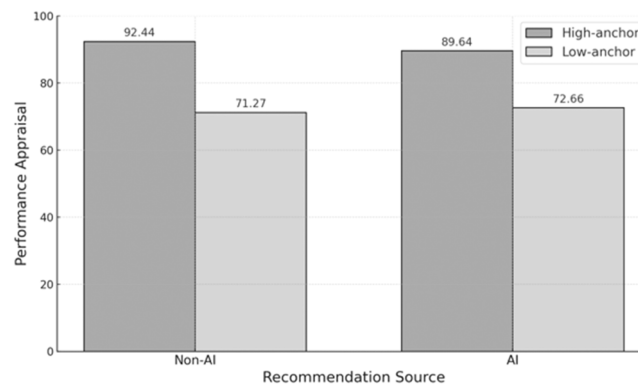
Now, indicate how would you assess your subordinate on a scale from 0 to 100: **AI Recommendation Scenarios:**

Imagine that you have to assess this year's performance of a subordinate of yours. During this year, your subordinate met the majority of goals, had good interpersonal skills with his colleagues, and showed moderate creativity in proposing new ideas for the improvement of the services.

Your company does a performance review every year. The previous year, an AI rating system called PeopleX recommended you assign your subordinate a performance rating of 51/100[91/100]. You have to decide whether to assign a rating lower or higher than 51/100[91/100].

Now, indicate how would you assess your subordinate on a scale from 0 to 100:

Appendix B. Mean Performance Appraisal Scores for AI and Non-AI Recommendations



Appendix C. Experiment 2 Adapted from Nagtegaal et al. (2020)

AI Recommendation, without CTO Scenarios:

Imagine that you have to assess this year's performance of a subordinate of yours. During this year, your subordinate met the majority of goals, had good interpersonal skills with his colleagues, and showed moderate creativity in proposing new ideas for the improvement of the services.

Your company does a performance review every year. The previous year, an AI rating system called PeopleX recommended you assign your subordinate a performance rating of 51/100[91/100]. You have to decide whether to assign a rating lower or higher than 51/100[91/100].

Now, indicate how would you assess your subordinate on a scale from 0 to 100:

AI Recommendation, with CTO Scenarios:

Imagine that you have to assess this year's performance of a subordinate of yours. During this year, your subordinate met the majority of goals, had good interpersonal skills with his colleagues, and showed moderate creativity in proposing new ideas for the improvement of the services.

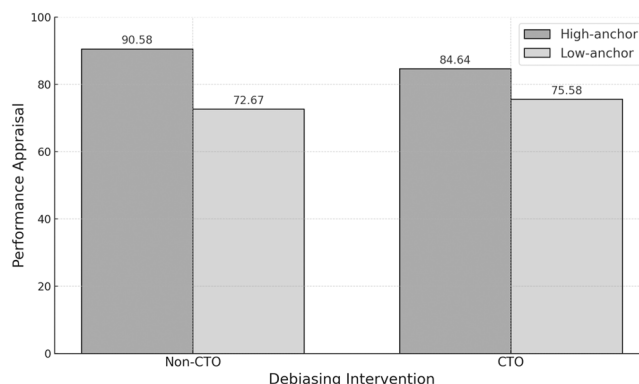
Your company does a performance review every year. The previous year, an AI rating system called PeopleX recommended you assign your subordinate a performance rating of 51/100[91/100]. You have to decide whether to assign a rating lower or higher than 51/100[91/100].

To make this decision, please first list two reasons why 51/100[91/100] might be too low[high] a rating for the employee's performance this year.

- Reason1 _____
- Reason2 _____

Now, indicate how would you assess your subordinate on a scale from 0 to 100:

Appendix D. Mean Performance Appraisal Scores with and without CTO Strategy



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